A36: are the λ = 1/4 forms in S(ρ_s) Hecke-character (dihedral) forms?

For: Arvin Hampton (539 Labs) · 2026-10-22, America/Chicago (CDT). `SF/` = `/workspace/quantum-comp/reports/s_family_paths/`. New files only, in `SF/A36_hecke_character/`.

Ground rules.

- CORE_FREEZE is respected: H_s is not redefined, no certifier code was touched or run, and the s31 mm sector was not touched.
- Jobs ran under `nice` with OMP_NUM_THREADS=1 and `ulimit -v` = 3 GB. Peak RSS was about 0.7 GB.
- Nothing was pushed or sent.

Preregistration. `prereg.md` was written before any computation. Its SHA-256 is `4810c430cf50da096dc701f305d49f87dd397929151d030c971dca9383b3815b` (`prereg.sha256`), and it is unchanged.

Categories. Cat A means proved, or exact (integer, rational, or exact mod ℓ) relative to the named hypotheses. Cat B means numerical or post hoc. The two are kept apart below.

Verdict per s

s	K_s	Cl(O), O = Z[√n]	dim S(ρ_s)	forms identified as K_s-dihedral	verdict
9	Q(√7)	Z/2	1	1: a quadratic ray-class character	**all dihedral**
11	Q(√11)	Z/4	2	2: a ring-class character of order 4, and a ray-class character	**all dihedral**
23	Q(√483)	$(Z/2)^2$	3	3: quadratic ray-class characters	**all dihedral**
25	Q(√23)	Z/6	3	3: one ring-class character (order 3 or 6) and two ray-class characters	**all dihedral**
27	Q(√3)	Z/6	3 or 4	3: same pattern as s = 25	**some dihedral** (3 forms); a possible 4th form is undetermined
29	Q(√87)	Z/6	3 or 4	3: same pattern as s = 25	**some dihedral** (3 forms); a possible 4th form is undetermined

Here "all dihedral" means that every form in S(ρ_s) is a linear combination of theta series θ_ψ of finite-order Hecke characters ψ of K_s with $\psi \neq \psi \circ \sigma$. These are the same kind of object as the Q(√6) form behind σ(q). However, none of the fields here is Q(√6).

1. Structural result (Cat A, relative to TB/TB4 and the classical Hecke–Maass theorem)

Lemma. Write $Q(x) = x_1^2 + s\,x_1x_2 + (s/2)x_2^2$, where $n = s(s-2)$. Then $Q(x) = N(x_1 + \omega x_2)$, with $\omega = (s+\sqrt{n})/2$ in $K_s = Q(\sqrt{n})$.

Why. 2Q is primitive of discriminant 4n. So Λ = Z + Zω is an invertible ideal of the order O = Z[√n].

Consequence for u. Each component of the lift is a_{α,β}(m) = ½ Σ_{ν∈α+Λ, Q(ν)=m, mod ε²} e(B(ν,β)), where ε = (s−1)+√n. Every component of u is therefore a Hecke theta series of a function on $D^{-1}Λ/Λ$ that is invariant under ε².

- For s = 9, …, 29, ε is the fundamental unit of O, of norm +1 (continued fraction, `out/classdata.json`).
- Expanding the weight function in characters of the group G_0 = I_D(O)/{(ν) : ν ≡ 1 mod D, N(ν) > 0} writes each component as a finite combination of θ_ψ(k·).
- The constraint "invariant along unit orbits, no |ν/ν′|^{it} factor" is exactly the condition for spectral parameter 0, that is λ = 1/4.
- Characters with ψ = ψ∘σ give Eisenstein series. Cusp parts involve only ψ ≠ ψ∘σ.

Closure under the operations used. P_σ is a constant matrix and T′_p (p ≡ 1 mod Tlevel) acts diagonally on θ_ψ, so the space of such combinations is preserved by both.

Where the lower bounds come from. At s = 9, 11, 23 and 25, every certified lower bound is built only from u:

- s = 9: u_cusp;
- s = 23: P_σu (the sector lemma);
- s = 11 and 25: P_σu and T′_pP_σu (`A9/certs/hecke_lower_s*.json`).

At each of these s the lower bound equals the certified upper bound. Hence S(ρ_s) is exactly the lift-derived span, and so (H2) S(ρ_s) is K_s-dihedral at s = 9, 11, 23 and 25.

s = 27 and 29. The three certified forms are dihedral by the same argument. A fourth form, if one exists, is not in the Hecke span of u (A34 §3.3), so nothing is known about it.

2. Exact class data, T0 (Cat A, unconditional)

Method (`scripts/classdata.py`, `out/classdata.json`):

- Gauss reduction cycles and composition give Cl(O).
- The continued fraction of √n gives the fundamental unit.
- (O/DO)^× is enumerated exhaustively.
- $|G_0| = h\cdot|(O/DO)^\times| \,/\, |\langle -1, \varepsilon\rangle \bmod D|$.

No PARI or GRH input is used.

s	D	$h^+(4n)$	h(4n)	wide Cl		ring-class cusp pairs (h − |Cl[2]|)/2	|(O/DO)^×|	exp	$\|G_0\|$
9	126	4	2	Z/2	0		4536	126	36
11	198	8	4	Z/4	1		11880	990	120
23	966	8	4	$(Z/2)^2$	0		255024	10626	528
25	1150	12	6	Z/6	2		506000	25300	1320
27	1350	12	6	Z/6	2		486000	2700	1080
29	1566	12	6	Z/6	2		789264	21924	1512

Candidate pool. The candidates are the characters of G_0 with ψ ≠ ψ∘σ, taken up to ψ ~ ψ∘σ. Their number is $(|G_0| - |G_0^\sigma|)/2$. I did not compute $|G_0^\sigma|$, so this pool size is not given exactly.

Ring-class part. The ring-class part of the pool is given exactly in the table. It does **not** match the

dims (0, 1, 0, 2 against 1, 2, 3, 3). So the dims are not counted by ring-class characters alone, and ray-class characters of modulus dividing D are needed.

3. Exact Hecke eigenvalues, T1–T3 (Cat A as exact mod-ℓ identities)

Method (`scripts/hecke_modl.py`, `logs/run_s*.out`, `out/hecke_modl_s*.json`):

- The lift coefficients (from `A34/scripts/coeffsS.py`, imported read-only) are reduced exactly modulo a prime $\ell \approx 10^{12}$ with $\ell \equiv 1 \bmod \mathrm{lcm}(D^2$, phase denominators, E), where E = exp((O/DO)^×)·h.
- T′_p is computed for 8 split primes p ≡ 1 mod Tlevel.
- Rank-1 sectors without an Eisenstein part give λ_p directly.
- Rank-1 cusp sectors with an Eisenstein part, and rank-2 sectors, are solved using second-order coefficients $a(p_0^2 m)$. In the rank-2 case the 2×2 matrix of $T'_{\{p_0\}}$ was verified exactly on two extra columns at s = 25, 27 and 29.

Test. Is every λ_p of the form $\zeta + \zeta^{-1}$ with ζ^E = 1, i.e. $a_p = \psi(\mathfrak{p}) + \psi(\bar{\mathfrak{p}})$? The test is non-trivial: a random residue passes with probability about $E/(2\ell) \le 10^{-7}$ per prime.

Results. All 12 eigen-systems pass at all 8 primes. In the table, the entries are the orders of $\psi(\mathfrak{p})$ prime by prime, and "cl" is the wide class of $\mathfrak{p}$ in Cl(O), where 0 is principal.

s	sector	cl of $\mathfrak{p}$ (8 primes)	order of $\psi(\mathfrak{p})$	ring-class test (H4)
9	(3−,7−)	0 0 0 0 0 0 0 0	2 2 2 2 2 2 2 1	ray (λ = −2 at principal $\mathfrak{p}$)
11	(3+,11+)	3 3 3 0 0 0 0 0	2 2 2 1 1 1 1 1	**ring-class**, order-4 character (class 3 has order 2)
11	(3−,11−)	3 3 3 0 0 0 0 0	4 4 4 2 2 1 2 2	ray
23	(3+,7−,23−)	all 0	2 1 2 2 2 1 1 1	ray, quadratic
23	(3−,7+,23−)	all 0	1 2 1 1 1 2 2 1	ray, quadratic
23	(3−,7−,23+)	all 0	2 2 2 2 2 2 2 1	ray, quadratic
25	(5+,23+)	2 0 2 2 2 1 2 1	3 1 3 3 3 3 3 3	**ring-class** (order 3 or 6)
25	(5−,23−) #1	same	6 2 6 3 3 6 6 6	ray
25	(5−,23−) #2	same	2 2 2 1 1 2 2 2	ray, quadratic
27	(3+,5+)	3 3 0 4 4 3 3 3	3 3 1 3 3 3 3 3	**ring-class** (order 3 or 6)
27	(3−,5−)	same	1 1 2 1 1 1	ray, quadratic

s	sector	cl of $\mathfrak{p}$ (8 primes)	order of $\psi(\mathfrak{p})$	ring-class test (H4)
	#1		1 1	
27	(3−,5−) #2	same	3 3 2 3 3 3 3 3	ray
29	(3+,29+)	1 2 1 1 2 1 2 2	3 (all)	consistent with ring-class; there is no principal prime in the sample, so this is not a full test
29	(3−,29−) #1	same	1 1 1 1 1 1 1 2	ray, quadratic ($\lambda = -2$ on a class of order 3)
29	(3−,29−) #2	same	3 3 3 3 3 3 3 6	ray

Notes.

- **What is exact.** Each λ_p is computed exactly mod ℓ. Its membership in $\{\zeta + \zeta^{-1}\}$ is exact mod ℓ. Char-0 dihedrality itself comes from §1, and the mod-ℓ test is an independent exact consistency check of it.
- **Eigen-systems are distinct.** Within each s the systems differ, which matches the certified ranks.
- **s = 23 (Cat B observation).** The three quadratic systems multiply pointwise, $\psi_1\psi_2 = \psi_3$, on all 8 primes. They behave like the three non-trivial characters of a Klein four-group, as $\mathrm{Cl}(O_{4\cdot 483}) \cong (\mathbb{Z}/2)^2$ does. They are not ring-class characters, however, since every one of these $\mathfrak{p}$ is principal.
- **Klein-four image.** The quadratic systems (s = 9, 23, and one each at 25, 27, 29) have $a_p \in \{\pm 2, 0\}$, so their projective image is the Klein four-group. Such forms are dihedral with respect to three quadratic fields, and for even forms these fields are real. I did not identify the other two fields.
- **s = 25.** The second ring-class pair of $\mathrm{Cl} \cong \mathbb{Z}/6$ does not occur in $S(\rho_{25})$. The 3 certified forms are the three systems above, and a form with the other ring-class system would be a fourth independent eigenform. (Its system agrees with the (5+,23+) one on the 8 sampled primes, so the sample cannot tell which of the two ring-class pairs is the one present.)

4. Class-number formula, H5

Preregistered criterion: FAIL. No count of characters derived from the data reproduces the dims. The visible systems split as (ring, ray) = (0,1), (1,1), (0,3), (1,2), and there is no derived rule for which ray-class characters land in ρ_s.

Cat B, post hoc and not proved. The empirical formula **$\dim S(\rho_s) = h(4n) - |\mathrm{Cl}(4n)^2| = h\cdot(1 - 1/|\mathrm{Cl}[2]|)$** gives 1, 2, 3, 3 at s = 9, 11, 23, 25. This is exactly right at all four. It is A2's predictor, with the [2,2] case corrected: $4 - 1 = 3$ at s = 23.

- At s = 27 and 29 it predicts 3, because $\mathrm{Cl} \cong \mathbb{Z}/6$.
- It was fitted after seeing the data and has no derivation.
- It does **not** decide s = 27 or 29.

Why s = 27 and 29 stay open. Even a proved "all dihedral" formula would need the unknown fourth form to be dihedral too, and the lift's Hecke span cannot see that form.

5. What is not claimed

- Dihedrality of a hypothetical fourth form at s = 27 or 29, or that such a form exists or does not exist.
- Any statement about other real quadratic fields beyond K_s, apart from the Klein-four remark above.
- The exact size of the full candidate pool, since $|G_0^{\sigma}|$ was not computed.

- No TPC/RH claims.

Files (all new)

- `prereg.md`, `prereg.sha256`, `CHECKPOINT.md`
- `scripts/classdata.py`, `scripts/hecke_modl.py`
- `out/classdata.json`, `out/hecke_modl_s{9,11,23,25,27,29}.json`
- `logs/run_s*.out`, `logs/hecke_modl_s*.log`
- `REPORT.md`, `REPORT.html`, `REPORT.pdf`